

Precept 3

Goals

- Prove the tower property and see its link to the Law of Total Probabilities or that of Expectations
- Use independent events to calculate probabilities of more complex events
- Use linearity of expectation

Reminder (Some useful formulas)

- **Factorials** : $\forall n \in \mathbb{N}^*, n! = n \times (n-1) \times \cdots \times 1, \quad 0! = 1$
- **Permutations** : Let P_k^n denote the number of ordered arrangements of k objects from a set of n objects.

$$P_k^n = n \times (n-1) \times (n-k+1) = \frac{n!}{(n-k)!}, \quad 0 \leq k \leq n$$

- **Combinations** : Let C_k^n denote the number of unordered arrangements of k objects from a set of n objects.

$$\begin{aligned} C_k^n &= \binom{n}{k} = \frac{n!}{k!(n-k)!}, \quad 0 \leq k \leq n \\ \binom{n}{r} &= \binom{n}{n-r}, \quad 0 \leq r \leq n \\ \binom{n+1}{r} &= \binom{n}{r} + \binom{n}{r-1}, \quad 1 \leq r \leq n \end{aligned}$$

- **Geometric series and derivatives formulas** :

Master formula, from which many other similar formulas follow immediately:

$$1 + r + r^2 + r^3 + \cdots + r^n = \sum_{i=0}^n r^i = \frac{1 - r^{n+1}}{1 - r}, \quad \text{for any } -1 < r < 1. \quad (1)$$

From there, we can get infinite series by letting $n \rightarrow \infty$ (noting that $r^n \rightarrow 0$):

$$\sum_{i=0}^{\infty} r^i = \frac{1}{1 - r}.$$

And we can of course compute other (finite or infinite) sums, that are the ones above from which we subtract some initial terms and/or multiply by some constant a . For example:

$$ar + ar^2 + ar^3 + \dots + ar^n = a \sum_{i=1}^n r^i = a \left(\frac{1 - r^{n+1}}{1 - r} - 1 \right).$$

Let $g(r) = \sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}$ for $r \in (-1, 1)$, $a \in \mathbb{R}$. Taking derivatives on both sides, the first derivative of g with respect to r is :

$$g'(r) = \sum_{k=1}^{\infty} akr^{k-1} = \frac{a}{(1-r)^2}.$$

Taking the derivatives again, we obtain :

$$g''(r) = \sum_{k=2}^{\infty} ak(k-1)r^{k-2} = \frac{2a}{(1-r)^3}.$$

Exercises

Exercise 0 (Tower Property $\equiv$ Law of Iterated Expectations (LIE))

We first discuss a useful property that you will often use, which relates conditional expectations to unconditional expectations. Let $X : \Omega \rightarrow D$, $Y : \Omega \rightarrow D'$ be discrete random variables with a joint distribution $\mathbf{P}\{X = x | Y = y\}$ for $x \in D$ and $y \in D'$. Prove that $\mathbf{E}[X] = \mathbf{E}[\mathbf{E}[X | Y]]$.

- **Law of Total Expectation :**

$$\mathbf{E}[X] = \sum_{y \in D'} \mathbf{E}[X | Y = y] \mathbb{P}(Y = y)$$

- **Law of Total Probability :**

$$\mathbb{P}(X = x) = \sum_{y \in D'} \mathbb{P}(X = x, Y = y) = \sum_{y \in D'} \mathbb{P}(X = x | Y = y) \mathbb{P}(Y = y)$$

- **Alternative versions for the Law of Total Expectation & Probability**

Let $\{A_i\}_{i=1,2,\dots}$ be a **finite or countably infinite partition** of the sample space Ω , then for a random variable $X : \Omega \rightarrow D$ defined on the probability space, we have :

$$\mathbf{E}[X] = \sum_i \mathbf{E}[X | A_i] \mathbb{P}(A_i)$$

For $x \in D$:

$$\mathbb{P}(X = x) = \sum_i \mathbb{P}(\{X = x\} \cap A_i) = \sum_i \mathbb{P}(X = x | A_i) \mathbb{P}(A_i).$$

Exercise 1 (Miner problem)

A miner is trapped in a mine containing three doors. The first door leads to a tunnel that takes him to safety after two hours of travel. The second door leads to a tunnel that returns him to the mine after three hours of travel. The third door leads to a tunnel that returns him to his mine after five hours. Assuming that the miner is at all times equally likely to choose any of the doors, what is the expected length of time until the miner reaches safety?

Exercise 2 (Dice games)

Two very good friends, Alice and Bob play the following game: Alice throws two fair, independent dice. Whenever the sum of two dice is less than 11, she throws it again—independent from the past rolls. If the sum is 12, Alice wins; and loses if the sum is 11.

- a. What is the probability that Alice wins “by” the n^{th} roll?
- b. What is the probability that Alice wins eventually?

Exercise 3 (Collecting action figures) - *time-permitting*

A cereal company puts an action figure in each box of cereal they sell. There are n different types of action figures that can be obtained. Each newly obtained action figure is, independently, type i with probability p_i , $i = 1, \dots, n$. Find the expected number of distinct types obtained in a collection of k cereal boxes.